

SAKTHIVEL MADHU

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PROFESSIONAL SUMMARY

Backend Engineer with **3+ years** of experience building **large-scale, automation-first enterprise platforms** across procurement, finance, and HR, plus hands-on experience delivering **production LLM agents**. Expertise in **Java 17, Spring Boot, microservices, event-driven architecture**, and **high-throughput REST APIs on GCP**. Track record of measurable business impact: **92% reduction in manual data entry, 80% faster inventory onboarding**, and a **multi-agent AI system** automating invoice-to-GRN processing, with full ownership from system design through production.

PROFESSIONAL EXPERIENCE

• Twinleaves Retail Ecommerce India Pvt. Ltd.

Bangalore, India

Software Development Engineer

May 2024 – Present

- **Purchase AI Agent:** Architected an autonomous agent (Java, Google Agent Development Kit, Gemini 2.5) that automates the vendor invoice-to-GRN workflow end-to-end, replacing fully manual data entry. Built multimodal LLM document extraction to convert invoice images/PDFs into structured JSON without a third-party OCR service, and a deterministic 5-tier product-matching engine (barcode, vendor code, HSN, fuzzy match, brand) producing reproducible confidence scores for a human-in-the-loop review gate. Structured multi-agent orchestration (coordinator + specialized sub-agents via tool-calling) integrated with Vendor, PO, Inventory, and Elasticsearch catalog services via REST, with SSE streaming for live review checkpoints.
- **Purchase & ERP Platform:** Led full-lifecycle development of enterprise **Purchase Order Management** (PR, PO, GRN, billing, payments, returns, inter-company workflows). Engineered **Express Purchase** (combined PO + GRN + put-away), cutting inventory onboarding time by **80%**. Integrated **Doc-AI** to reduce manual data entry by **92%** and shipped government-compliant **E-Invoicing** for Malaysian regulations.
- **HRMS: Facial Recognition, Attendance & LOP:** Created an HR attendance platform with **facial recognition** check-in/out, **geo-fencing** for location-based policies, and automated **LOP calculation** from leave balances, shifts, and attendance gaps. Built real-time dashboards and payroll-linked reporting, eliminating manual HR effort across distributed teams.
- **Vendor Management System:** Developed a centralized vendor onboarding platform with automated **GST, PAN, and bank verification** via third-party integrations through **API Gateway**, and configurable **Terms of Trade** engines for item-level pricing and return-policy rules.
- **Finance Management System (Pallet Books):** Developed a scalable finance platform, modeled on tools like Zoho Books and Tally, for ledgers, purchase accounting, and real-time **reconciliation**, with **Elasticsearch**-powered ledger search. Migrated bank statement extraction to **Doc-AI**, reaching **95% accuracy** across formats.
- Established **microservices** (Java 17, Spring Boot) with high-throughput REST APIs; optimized **PostgreSQL** schemas; implemented fault-tolerant async workflows with **Pub/Sub**, retries, and idempotent APIs. Set up **CI/CD** (GitHub Actions, Jenkins), **Docker**, and **Elasticsearch**-powered logging on **GCP**. Served as technical owner across product, finance, ops, and HR teams; mentored junior engineers.

• Masai School

Remote

Full Stack Web Developer

Jul 2022 – Jul 2023

- Completed 2160+ hours of full-stack and backend training; shipped 3+ production-style apps including **Naukri** and **e-commerce** clones and a backend-driven **Auction Management System** with real-time bidding. Stack: Java, Spring Boot, Hibernate, MySQL, JavaScript.

• EFI Pvt. Ltd.

Bangalore, India

Associate Software Engineer

Oct 2020 – Mar 2021

- Delivered 3+ feature modules for the **Fiery** product (C++, SQL) across two stable production releases, partnering with QA and business stakeholders on requirements, defect resolution, and release validation.

EDUCATION

Sri Venkateshwara College of Engineering, Chennai, Tamil Nadu

Jul 2014 – May 2018

B.E. in Electronics & Communication Engineering

TECHNICAL SKILLS

Backend, Cloud & DevOps: Java 17, Spring Boot, Spring Security, Spring Cloud, REST APIs, Microservices, Hibernate/JPA, Event-Driven Architecture, API Gateway, GCP, Docker, CI/CD (GitHub Actions, Jenkins), Maven, Gradle, Linux, OpenAPI/Swagger

AI & LLM Agents: Google Agent Development Kit (ADK), Gemini 2.5 (Pro/Flash), Multi-Agent Orchestration, Tool-Calling, Multimodal Document Extraction, Doc-AI, Prompt Engineering, SSE Streaming

Databases, Search & Messaging: PostgreSQL, MySQL, MongoDB, BigQuery, Elasticsearch, Redis (Caching & Pub/Sub), GCP Pub/Sub, Async Processing, SQL Optimization

Security & Languages: JWT, OAuth 2.0, API Security, Python, JavaScript, Node.js, Flask, C++, Postman

Domain & Practices: Accounting Workflows, Reconciliation, Ledger Systems, Payroll, LOP Calculation, Facial Recognition, Geo-fencing, E-Invoicing, System Design, Distributed Systems, DSA, Agile/Scrum, Code Review

ACHIEVEMENTS & CERTIFICATIONS

- **Employee of the Quarter: Twinleaves:** Recognized for high-impact engineering contributions across ERP, HRMS, and Finance platforms
- Prompt Engineering for Developers (Java): DeepLearning.AI / ChatGPT
- Python Programming Certification: Apponix Training Center